HOMEWORK 8.4 · UNIT 8 · NC.8.G.4 · NC.8.G.3 · NC.8.F.4

Dilations Day 2

Targets: G.4-2, G.3-5, G.3-4, G.3-2, G.3-3, G.3-7, G.3-6, F.4-5, G.3-9, G.3-8

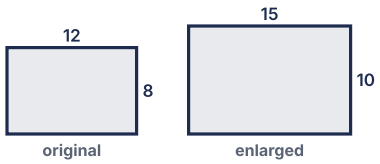
Name _____ Date _____ Block _____

Remember: Scale factor $k = \frac{\text{image length}}{\text{pre-image length}}$. $k > 1$: bigger. $k < 1$: smaller. To undo a dilation, divide by k. **Example:** $AB = 4$ and $A'B' = 10$, so $k = \frac{10}{4} = 2.5$.

SCORE 3.0 Approaching · the basics

Targets: G.4-2, G.3-5, G.3-4, G.3-2, G.3-3

1 A 12 by 8 photo is enlarged to 15 by 10. What is the scale factor? G.4-2



Ⓐ 1.25 Ⓑ 0.8 Ⓒ 2 Ⓓ 3

2 A(−2, 5) is dilated about the origin to A′(−6, 15). Find the scale factor. G.3-5

3 Dilate (8, −12) by $k = \frac{3}{4}$ about the origin. G.3-4

4 Translate P(−5, 3) by $(x + 7, y - 4)$. Find P′. REVIEW 8.1 G.3-2

Ⓐ (2, −1) Ⓑ (−12, 7)
Ⓒ (2, 7) Ⓓ (−5, 7)

5 Rotating (x, y) 180° about the origin gives $(-x, -y)$. REVIEW 8.2 G.3-3

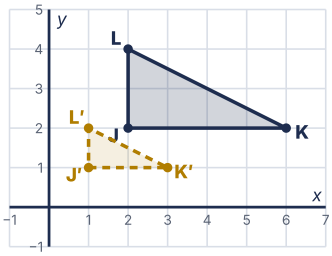
True False

SCORE 4.0 Proficient · on target Targets: G.3-7, G.3-6, F.4-5

6 **Zoe** made a mistake. Circle it. Explain it. Fix it. G.3-7

ZOE'S WORK
Dilate (5, 2) by $k = 4$.
(5 + 4, 2 + 4) = (9, 6)

7 Which single transformation maps △JKL onto △J'K'L'? G.3-6



(A) Dilation, $k = \frac{1}{2}$
(B) Dilation, $k = 2$
(C) Translation 1 left, 1 down
(D) Translation 3 left, 1 down

8 Segment AB has A(1, 2) and B(3, 6). It is dilated by $k = 3$ to A′(3, 6) and B′(9, 18). Find the slope of AB and the slope of A′B′. REVIEW 6.5 F.4-5

Hint: slope = $\frac{y_2 - y_1}{x_2 - x_1}$

SCORE 5.0 Mastery · go deeper Targets: G.3-9, G.3-8

9 A square has 2-inch sides. It is dilated by $k = 5$. Find the new side length and the new area. How many times bigger is the area? G.3-9

The new side is _____ . The area is _____ times bigger because _____ .

10 Point (4, 6) is reflected over the *y*-axis. Then it is dilated by $k = \frac{1}{2}$. Find the final point. Is a figure moved this way congruent or similar? Explain. G.3-8

The final point is _____ . The figure is _____ because _____ .